

SOLUTIONS

- ① One-dimensional wave equation for the string, $0 \leq x \leq l$:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0 \quad (1)$$

with boundary conditions $u(0, t) = 0$
 $u_x(l, t) = 0$

[fixed end at $x=0$; "free" end at $x=l$]

Seek solution in the form:

$$u(x, t) = v(x) q(t) \quad (2)$$

Substituting into (1):

$$v(x) \frac{\partial^2 q}{\partial t^2} - q(t) c^2 \frac{\partial^2 v}{\partial x^2} = 0,$$

dividing by $v(x)q(t)$ and replacing partial derivatives by total ($\partial \rightarrow d$):

$$\frac{1}{q(t)} \frac{d^2 q}{dt^2} - \frac{c^2}{v(x)} \frac{d^2 v}{dx^2} = 0 \quad (3)$$

function of
t only

function of
x only

The two terms in (3) must be constants, i.e., denoting the constant $-\omega^2$:

$$\frac{1}{q(t)} \frac{d^2 q}{dt^2} = -\omega^2 \quad (4)$$

[We will see below that the separation constant must be negative.] Eq. (4) gives: $q'' + \omega^2 q = 0$,

So, The general solution of (4) is

(2)

$$q(t) = C_1 \cos \omega t + C_2 \sin \omega t,$$

which can also be written as

$$q(t) = A \cos(\omega t + \alpha), \text{ where } \alpha \text{ is}$$

the initial phase.

Turning to the coordinate part of Eq. (3) now:

$$\frac{c^2}{v(x)} \frac{d^2 v}{dx^2} = -\omega^2$$

$$\frac{d^2 v}{dx^2} = -\frac{\omega^2}{c^2} v(x);$$

Denoting $\frac{\omega}{c} \equiv k$, we have:

$$v'' + k^2 v = 0$$

(5)

The general solution of this equation is:

$$v(x) = C_1 \cos kx + C_2 \sin kx.$$

Using the 1st boundary condition, $u(0, t) = 0$,
we must have $v(0) = 0$, hence: $0 = C_1 \cdot 1 + C_2 \cdot 0$,

and $C_1 = 0$.

Therefore $v(x) = C \sin kx$, where we have
renamed $C_2 \rightarrow C$.

Using the second boundary condition: $v'(l) = 0$:

$$v'(x) = Ck \cos kx$$

$$u_x(l, t) = 0$$

$$Ck \cos kl = 0$$

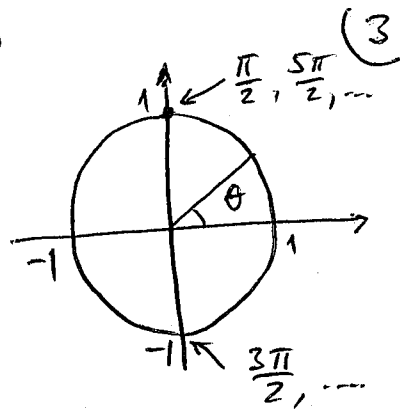
To have a nonzero solution, we require

$$\underline{\cos kl = 0}.$$

For positive θ , $\cos \theta$ vanishes

at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots$,

$$\text{i.e. } \theta = \frac{\pi}{2} + n\pi, \quad n = 0, 1, 2, \dots$$



Hence
$$kl = \pi \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

$$k = \frac{\pi}{l} \left(n + \frac{1}{2} \right)$$

$$v_n(x) = C \sin \frac{(n + \frac{1}{2})\pi x}{l} \quad (6)$$

The corresponding frequencies are

$$\omega = kc = \frac{\pi c (n + \frac{1}{2})}{l} \equiv \omega_n$$

Combining $v(x)$, Eq. (6), with $q(t)$, we have:

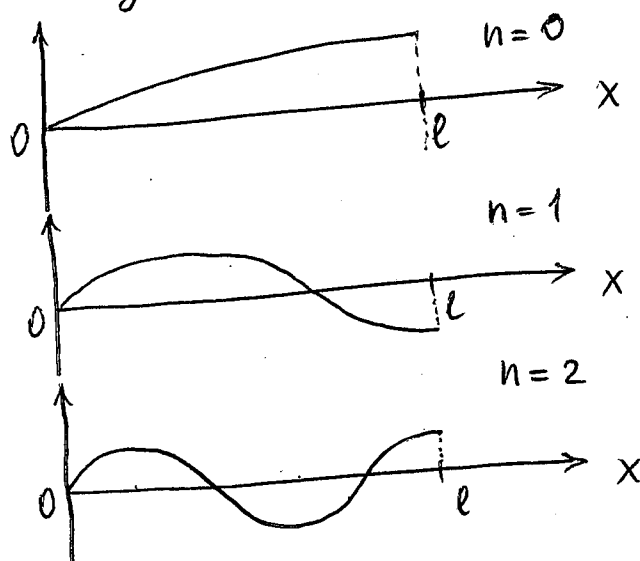
$$u(x, t) = A \sin \frac{(n + \frac{1}{2})\pi x}{l} \cos(\omega_n t + \alpha)$$

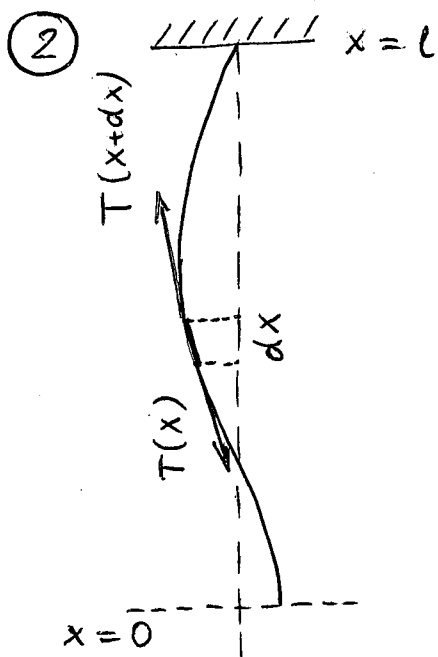
where A is an arbitrary constant.

These solutions describe harmonic vibrations with frequency ω_n . They can be visualised by plotting $v_n(x)$.

In these solutions the phase of the sin function increases from 0 at $x = 0$ to $(n + \frac{1}{2})\pi$

at $x = l$. The number of nodes between $x = 0$ and $x = l$ is n .





(a) Considering the segment of the rope of length dx :

its mass is ρdx ,

Forces acting on it at the two ends: $T(x+dx)$ and $T(x)$.

Taking their components in the direction perpendicular to x , we obtain the total force acting on dx segment in this

direction:

$$T(x+dx) \sin \theta(x+dx) - T(x) \sin \theta(x)$$

$$= \frac{\partial}{\partial x} (T(x) \sin \theta(x)) dx, \quad (1)$$

where $\theta(x)$ is the angle the rope makes with the x direction.

If the perpendicular displacement of the rope is $u(x,t)$, then, for small displacements and "nearly straight" rope, $\theta \ll 1$,

$$\sin \theta \approx \tan \theta = \frac{\partial u}{\partial x} \quad (2)$$

The acceleration of the segment is $\frac{\partial^2 u}{\partial t^2}$, hence by Newton's 2nd law:

$$\rho dx \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x} \left(T(x) \frac{\partial u}{\partial x} \right) dx \quad (3)$$

The tension in the rope at point x is equal to the weight of the rope below, i.e. $T = \rho x g$.

Substituting this into (3) we obtain:

$$\rho \frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x} \left(\rho x g \frac{\partial u}{\partial x} \right) \quad (5)$$

Cancelling ρ and shifting the term from RHS to LHS we have:

$$\frac{\partial^2 u}{\partial t^2} - g \frac{\partial}{\partial x} \left(x \frac{\partial u}{\partial x} \right) = 0, \quad (4)$$

as required.

(b) Seek solution of (4) in the form:

$$u(x, t) = v(x) q(t). \quad (5)$$

Substituting (5) into (4):

$$v(x) \frac{\partial^2 q}{\partial t^2} - q(t) g \frac{\partial}{\partial x} \left(x \frac{\partial v}{\partial x} \right) = 0$$

Dividing through by $v(x) q(t)$:

$$\underbrace{\frac{1}{q(t)} \frac{d^2 q}{dt^2}}_{\text{function of } t} - \underbrace{\frac{g}{v(x)} \frac{d}{dx} \left(x \frac{dv}{dx} \right)}_{\text{function of } x} = 0$$

The two terms must be constants. Choosing the separation constant as $-\omega^2$, we have for the temporal part:

$$\frac{1}{q(t)} q'' = -\omega^2$$

$$q'' + \omega^2 q = 0$$

General solution:

$$q(t) = C_1 \cos \omega t + C_2 \sin \omega t$$

$$\text{or } q(t) = A \cos(\omega t + \alpha),$$

where α is the initial phase.

Turning to the spatial, coordinate part now: (6)

$$\frac{g}{v(x)} \frac{d}{dx} \left(x \frac{dv}{dx} \right) = -\omega^2$$

$$g \frac{d}{dx} \left(x \frac{dv}{dx} \right) + \omega^2 v(x) = 0 \quad \left. \vphantom{\frac{d}{dx} \left(x \frac{dv}{dx} \right)} \right\} \text{Using product rule}$$

$$g \frac{dv}{dx} + g x \frac{d^2 v}{dx^2} + \omega^2 v(x) = 0 \quad : \text{ divide by } g$$

$$x \frac{d^2 v}{dx^2} + \frac{dv}{dx} + \frac{\omega^2}{g} v = 0, \quad (6)$$

as required.

(c) New variable $\xi = \alpha \sqrt{x}$, i.e. $x = \xi^2 / \alpha^2$.

We need to express the derivatives in Eq. (6) in terms of derivatives with respect to ξ .

$$\frac{dv}{dx} = \frac{dv}{d\xi} \frac{d\xi}{dx} = \frac{dv}{d\xi} \cdot \frac{\alpha}{2\sqrt{x}}$$

$$\frac{d^2 v}{dx^2} = \frac{d}{dx} \left(\frac{dv}{dx} \right) = \frac{d}{dx} \left(\frac{dv}{d\xi} \cdot \frac{\alpha}{2\sqrt{x}} \right)$$

Using chain rule and then product and chain rule for the 2nd derivative

$$= \frac{d}{dx} \left(\frac{dv}{d\xi} \right) \cdot \frac{\alpha}{2\sqrt{x}} + \frac{dv}{d\xi} \frac{d}{dx} \left(\frac{\alpha}{2\sqrt{x}} \right)$$

$$= \frac{d^2 v}{d\xi^2} \cdot \frac{d\xi}{dx} \cdot \frac{\alpha}{2\sqrt{x}} + \frac{dv}{d\xi} \cdot \left(-\frac{\alpha}{4x^{3/2}} \right)$$

$$= \frac{d^2 v}{d\xi^2} \cdot \frac{\alpha^2}{4x} - \frac{\alpha}{4x^{3/2}} \frac{dv}{d\xi}$$

Substituting the derivatives into (6) and expressing $\sqrt{x}$, x , $x^{3/2}$ in terms of ξ , we have:

$$\frac{\xi^2}{\alpha^2} \left(\frac{d^2 v}{d\xi^2} \cdot \frac{\alpha^2}{4\xi^2/\alpha^2} - \frac{\alpha \alpha^3}{4\xi^3} \frac{dv}{d\xi} \right) + \frac{dv}{d\xi} \cdot \frac{\alpha \alpha}{2\xi} + \frac{\omega^2}{g} v = 0$$

$$\frac{\alpha^2}{4} \frac{d^2 v}{d\xi^2} - \frac{\alpha^2}{4\xi} \frac{dv}{d\xi} + \frac{\alpha^2}{2\xi} \frac{dv}{d\xi} + \frac{\omega^2}{g} v = 0 \quad \left\} \times \frac{4}{\alpha^2} \quad (7)$$

$$\frac{d^2 v}{d\xi^2} + \frac{1}{\xi} \frac{dv}{d\xi} + \frac{4\omega^2}{\alpha^2 g} v = 0$$

Choosing $\frac{4\omega^2}{\alpha^2 g} = 1 \Leftrightarrow \alpha = \frac{2\omega}{\sqrt{g}}$,
we have:

$$\frac{d^2 v}{d\xi^2} + \frac{1}{\xi} \frac{dv}{d\xi} + v = 0, \quad (7)$$

as required.

(d) Equation (7) is the Bessel equation for $m=0$.
Its solution finite at $x=0$ (bottom end
($\xi=0$) of the rope) is:

$$v = J_0(\xi) = J_0(\alpha\sqrt{x})$$

$$\Rightarrow v = A J_0\left(\frac{2\omega\sqrt{x}}{\sqrt{g}}\right) \quad \left\{ \begin{array}{l} \text{We can multiply the} \\ \text{solution of a} \\ \text{homogeneous equation} \\ \text{by an arbitrary const.} \end{array} \right.$$

At the top end, $x=l$, the string is fixed, i.e.

$$u(l, t) = 0, \text{ hence, } v(l) = 0:$$

$$J_0\left(\frac{2\omega\sqrt{l}}{\sqrt{g}}\right) = 0$$

Therefore, $\frac{2\omega\sqrt{l}}{\sqrt{g}} = z_{0,n}$, where $z_{0,n}$ is the n th root of $J_0(z)$. So, the frequency is given by

$$\omega = \frac{z_{0,n}\sqrt{g}}{2\sqrt{l}} \equiv \omega_n,$$

$$\text{or } \omega_n = \frac{z_{0,n}}{2} \sqrt{\frac{g}{l}}$$

Substituting this ω value into $v(x)$ we have: (8)

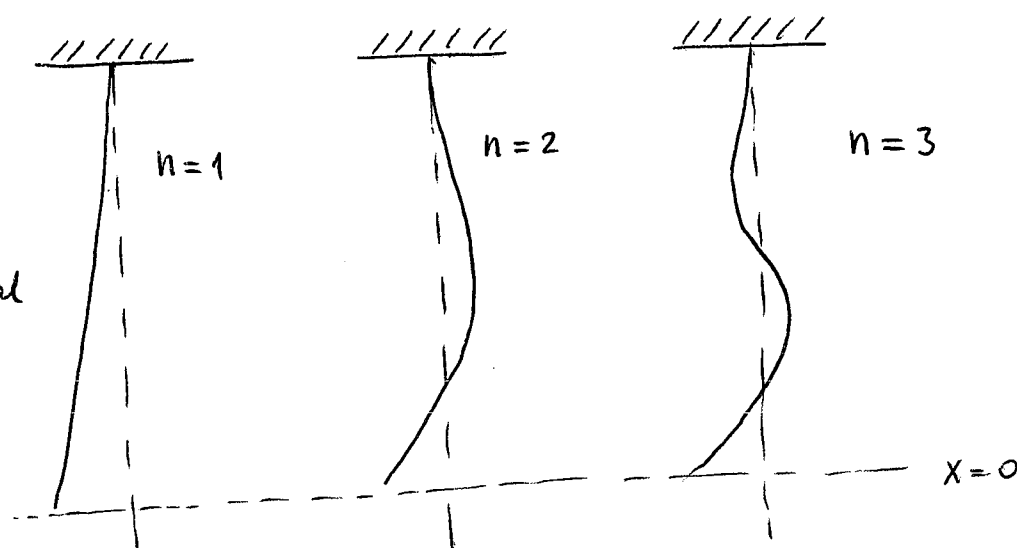
$$v(x) = A J_0 \left(z_{0,n} \sqrt{\frac{x}{l}} \right).$$

(e) Combining the above expression with $q(t)$ obtained in part (b), we find:

$$u(x,t) = A J_0 \left(z_{0,n} \sqrt{\frac{x}{l}} \right) \cos(\omega_n t + \alpha).$$

This solution describes harmonic vibrations of the hanging rope. They can be visualised by sketching the shape of the rope for a fixed t .

Higher n obviously correspond to higher vibrational frequencies.



③ (a)
$$\frac{\partial u}{\partial t} - K \frac{\partial^2 u}{\partial x^2} = 0 \quad (1)$$

For a time independent solution $u(x,t) = u_s(x)$

$\frac{\partial u_s}{\partial t} = 0$, hence $u_s(x)$ satisfies:

$$-K \frac{d^2 u_s}{dx^2} = 0 \quad (2)$$

[Note: u_s depends on x only, hence we use the total derivative, not partial.]

The solution of (2), or equivalently, $u_s'' = 0$ is a linear function:

$$u_s(x) = C_1 x + C_2.$$

Using the boundary conditions: $u_s(0) = T_1$, $u_s(l) = T_2$ (9)

$$C_1 \cdot 0 + C_2 = T_1 \Rightarrow C_2 = T_1$$

$$C_1 l + C_2 = T_2 \Rightarrow C_1 = (T_2 - T_1)/l$$

$$\Rightarrow \underline{u_s(x) = T_1 + \frac{T_2 - T_1}{l} x}, \text{ as required.}$$

(b) Suppose, $u_0(x, t)$ is a solution of (1) with boundary conditions $u_0(0, t) = u_0(l, t) = 0$.

Consider: $u(x, t) = u_0(x, t) + u_s(x)$.

Substituting into (1) (left-hand side):

$$\begin{aligned} & \frac{\partial}{\partial t} (u_0(x, t) + u_s(x)) - K \frac{\partial^2}{\partial x^2} (u_0(x, t) + u_s(x)) \\ &= \underbrace{\frac{\partial u_0}{\partial t} - K \frac{\partial^2 u_0}{\partial x^2}}_{\substack{= \\ 0}} - \underbrace{K \frac{\partial^2 u_s}{\partial x^2}}_{\substack{= \\ 0}} = 0. \end{aligned}$$

Hence $u = u_0 + u_s$ is a solution of the heat eq'n.

At the boundaries: $u(0, t) = u_0(0, t) + u_s(0) = T_1$,
 $u(l, t) = u_0(l, t) + u_s(l) = T_2$.

(c) We know from Sec. 2.4 that $u_0(x, t)$ can be constructed as a sum:

$$u_0(x, t) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} e^{-(n^2\pi^2/l^2)Kt}.$$

Hence, the solution that satisfies the boundary conditions $u(0, t) = T_1$, $u(l, t) = T_2$, as we have seen in part (b), can be obtained as the sum:

$$u(x, t) = \underbrace{T_1 + \frac{T_2 - T_1}{l} x}_{u_s(x)} + \underbrace{\sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} e^{-(n^2\pi^2/l^2)Kt}}_{u_0(x, t)}$$

To satisfy the initial condition $u(x, 0) = f(x)$, (10)
we require:

$$T_1 + \frac{T_2 - T_1}{l} x + \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} = f(x)$$

Note:
 $e^{-(n^2\pi^2/l^2)kt}$
 equals 1
 for $t=0$.

$$\sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{l} = f(x) - T_1 - \frac{T_2 - T_1}{l} x$$

Multiplying by $\sin \frac{m\pi x}{l}$ and integrating over x between 0 and l , we use the first of the integrals in Examples (Q.2) and see that only one term, with $n=m$, gives a nonzero contribution.

Hence:

$$B_m \frac{l}{2} = \int_0^l \left(f(x) - T_1 - \frac{T_2 - T_1}{l} x \right) \sin \frac{m\pi x}{l} dx$$

$$\Rightarrow B_m = \frac{2}{l} \int_0^l \left[f(x) - T_1 - \frac{T_2 - T_1}{l} x \right] \sin \frac{m\pi x}{l} dx,$$

as required.

[B_n is given by the same expression with $\sin \frac{n\pi x}{l}$ under the integral on the right-hand side.]

④ Two dimensional wave equation in Cartesian coordinates x and y is

$$\frac{\partial^2 u}{\partial t^2} - c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) = 0, \quad (1)$$

for $u(x, y, t)$.

Separating the variables, let us seek solution in the form:

$$u(x, y, t) = v(x) w(y) q(t). \quad (2)$$

Substituting into (1), we have:

$$v(x)w(y) \frac{d^2 q}{dt^2} - c^2 w(y)q(t) \frac{d^2 v}{dx^2} - c^2 v(x)q(t) \frac{d^2 w}{dy^2} = 0 \quad (11)$$

Dividing by $v(x)w(y)q(t)$, we get:

$$\underbrace{\frac{1}{q} q''}_{\text{depends on } t} - c^2 \underbrace{\frac{1}{v} v''}_{\text{depends on } x} - c^2 \underbrace{\frac{1}{w} w''}_{\text{depends on } y} = 0 \quad (3)$$

Hence, each of the terms must be a constant.

Let $\frac{1}{q} q'' = -\omega^2$ (negative separation constant).

$$q'' + \omega^2 q = 0$$

The general solution is

$$q(t) = C_1 \cos \omega t + C_2 \sin \omega t$$

or $q(t) = A \cos(\omega t + \phi)$.

With this choice of separation constant, (3)

becomes:

$$-\omega^2 - c^2 \frac{v''}{v} - c^2 \frac{w''}{w} = 0.$$

or $c^2 \frac{v''}{v} + c^2 \frac{w''}{w} + \omega^2 = 0$

or $\frac{v''}{v} + \frac{w''}{w} + k^2 = 0$, where $k = \frac{\omega}{c}$.

Let $\frac{v''}{v} = -k_1^2$, $\frac{w''}{w} = -k_2^2$, $\left. \begin{array}{l} \text{We choose} \\ \text{negative} \\ \text{constants here.} \\ \text{see below.} \end{array} \right\}$

so that $-k_1^2 - k_2^2 + k^2 = 0$. (4)

Equation for $v(x)$:

$$v'' + k_1^2 v = 0$$

General solution: $v(x) = A \cos k_1 x + B \sin k_1 x$.

The boundary conditions $u(0, y, t) = u(a, y, t) = 0$ applied to (2) mean: $v(0) = v(a) = 0$. (12)

Hence $A = 0$ and $v(x) = B \sin k_1 x$, and

$$B \sin k_1 a = 0$$

$$\Rightarrow k_1 a = n\pi, \quad n = 1, 2, \dots$$

$$\underline{k_1 = \frac{n\pi}{a}}$$

Similarly, the equation for $w(y)$ is

$$w'' + k_2 w = 0.$$

Its solution $w(y) = C \cos k_2 y + D \sin k_2 y$ satisfies the boundary conditions $u(x, 0, t) = u(x, b, t) = 0$, i.e. $w(0) = w(b) = 0$, only if $C = 0$,

$$w(y) = D \sin k_2 y, \quad \text{and} \quad \sin k_2 b = 0,$$

$$\text{hence} \quad k_2 b = m\pi, \quad m = 1, 2, \dots$$

$$\underline{k_2 = \frac{m\pi}{b}}$$

[Note that if we chose positive separation constants, e.g. $\frac{v''}{v} = k_1^2$, i.e. $v'' - k_1^2 v = 0$, the solution would be $v(x) = A e^{k_1 x} + B e^{-k_1 x}$, and it would not satisfy $v(0) = v(a) = 0$ for nonzero A or B , for any k_1 .]

Substituting k_1 and k_2 into (4): $k^2 = \frac{n^2 \pi^2}{a^2} + \frac{m^2 \pi^2}{b^2}$, and the frequency of harmonic vibrations is

$$\omega = ck = \pi c \sqrt{\frac{n^2}{a^2} + \frac{m^2}{b^2}} \equiv \omega_{nm}.$$

The corresponding solution is obtained by combining $q(t)$, $v(x)$ and $w(y)$ derived above into (2):

$$u(x, y, t) = \underbrace{A}_{\substack{\text{arbitrary} \\ \text{constant}}} \underbrace{\sin \frac{n\pi x}{a}}_{v(x)} \underbrace{\cos \frac{m\pi y}{b}}_{w(y)} \underbrace{\cos(\omega_{nm} t + \phi)}_{q(t)}.$$